Beyond-Earth: NASA Scientific Data & Dataset Inventory

Comprehensive Source-Code Audit of NASA Data Products, Cartographic Textures, and Space Science Models

Project Name: Beyond-Earth	**Repository: github.com/kazinayeem/Beyond-Earth**
Audit Date: October 9, 2026	**Audit Scope: Complete Source Code, Assets, Textures, Adapters & APIs**
Audit Status: Verified & Reconciled	**Primary Framework: Next.js 16 (App Router), Three.js, WebGL**

1. Executive Summary of Identified NASA Data

An exhaustive audit of the **Beyond-Earth** codebase was conducted across all source files, React components, Three.js 3D shaders, static assets in /public/textures/, TypeScript data models, adapters, Zod validation schemas, and internal API routes. The project integrates authentic NASA open data through four distinct tiers: **(1) Embedded NASA Cartographic Image Textures**: 2K and 4K satellite composites derived from MODIS Terra/Aqua, SRTM elevation models, and GOES meteorological imagery powering real-time 3D Earth rendering; **(2) Canonical Planetary Constants**: Real astrophysical measurements (mass, mean radius, surface gravity, escape velocity, exosphere mixing ratios, diurnal temperature swings, and radiation dosages) curated by NASA GSFC/NSSDC for the Moon, Mars, and Earth; **(3) Lunar Reconnaissance Orbiter (LRO) Scientific Payloads**: Verified physical engineering parameters (mass, power) and scientific phenomena for six authentic LRO instruments downlinked to the NASA Planetary Data System (PDS); and **(4) Historical Mission & Orbital Ephemeris Profiles**: Canonical NASA mission archives (Apollo 11, LRO, Mars 2020, OSIRIS-REx) and CelesTrak/NASA Tracking Network two-line orbital elements for active spacecraft (ISS, Hubble, Landsat 9, NOAA-20). All genuine scientific data are cleanly isolated from game-balanced progression mechanics.

2. Confirmed NASA Datasets and Data Products Table

The following table catalogues all genuine NASA data products confirmed to be actively used within the application.

#	Dataset / Product Name	Mission / Node Origin	Data Type	Delivery / Storage	Target Feature
1	**NASA Blue Marble 2K Surface Texture (earth_surface_2k.jpg)**	EOS Terra & Aqua / MODIS (GSFC)	Image (2K RGB)	Local Asset (512 KB JPG)	Title Hero & Mission Control 3D Earth
2	**NASA Blue Marble Ocean Specular Mask (earth_specular_2k.jpg)**	MODIS Land/Water Classification	Image (2K Mask)	Local Asset (223 KB JPG)	Three.js Ocean Sun Reflection Mask
3	**NASA SRTM Elevation Normal Map (earth_normal_2k.jpg)**	Shuttle Radar Topography Mission	Image (2K Normal)	Local Asset (336 KB JPG)	Topographic Surface Bump Lighting
4	**NASA/NOAA Meteorological Cloud Map (earth_clouds_4k.png)**	GOES / NOAA Weather Satellites	Image (4K Alpha)	Local Asset (5.03 MB PNG)	Concentric 3D Atmospheric Cloud Sphere
5	**NASA Planetary Fact Sheets (Moon, Mars, Earth)**	NASA GSFC / NSSDC Coordination	Scientific Constants	TypeScript Model + API Route	Mission Briefings & Data Explorer
6	**LRO LOLA Topography GDR (LRO-L-LOLA-3-RDR-V1.0)**	LRO / NASA PDS Geosciences Node	Laser Altimetry (DEM)	Structured Catalog + API	Laser Altimeter CAD & Synergies
7	**LRO LROC Camera Archive (LRO-L-LROC-2-EDR-V1.0)**	LRO / NASA PDS Imaging Node / ASU	Optical Imagery / EDR	Structured Catalog + API	LROC Camera Payload & Builder
8	**LRO Diviner Thermal Data (LRO-L-DLRE-4-RDR-V1.0)**	LRO / NASA PDS Geosciences / UCLA	Radiometer (Thermal)	Structured Catalog + API	Diviner Payload, Polar Cold Traps Event
9	**LRO Mini-RF SAR Polar Ice Data (LRO-L-MRFLRO-4-CDR-V1.0)**	LRO / NASA PDS Geosciences / JHU-APL	Synthetic Aperture Radar	Structured Catalog + API	Radar Sensor CAD & Subsurface Ice Event

10	**LRO LAMP UV Science Data (LRO-L-LAMP-2-EDR-V1.0)**	LRO / NASA PDS Atmospheres Node / SwRI	Far-UV Spectroscopy	Structured Catalog + API	LAMP UV Spectrograph Payload
11	**LRO CRaTER Radiation Archive (LRO-L-CRAT-2-EDR-V1.0)**	LRO / NASA PDS PPI Node / UNH	LET Dosimetry Spectra	Structured Catalog + API	CRaTER Detector & Solar Storm Crisis
12	**NASA Robotic Missions Master Catalog (NSSDCA)**	GSFC / JPL / NSSDCA Archives	Mission Profiles & TB	Structured Catalog + API	Robotic Missions Explorer Tab
13	**NASA Tracking Network / CelesTrak Satellite Ephemeris**	NASA / Space-Track / CelesTrak	TLE Orbital State Vectors	Hardcoded Physics Array	Hero 3D Satellite Tracking Simulation
14	**NASA Deep Space Network Standards (JPL D-16838)**	NASA JPL DSN Ground Operations	RF Carrier Protocols	Hardcoded Rules / State	Live DSN Beam Lock & Comm Drop Event
15	**NASA DONKI / NOAA SWPC Solar Flare Catalog**	CCMC DONKI / NOAA SWPC	Space Weather Proton Flux	Branching Event Logic	Class-X Coronal Mass Ejection Event

3. Detailed Dataset Breakdown & Specific Information Used

Each identified dataset was verified for authentic scientific origin, exact information content, and concrete data flow into the user-facing application features.

A. Earth Cartographic Texture Suite (NASA Earth Observatory / Blue Marble)

The project embeds four high-resolution equirectangular raster datasets inside public/textures/. These files originate from NASA's Earth Observing System (EOS) and meteorological satellite composites, distributed via the Three.js scientific graphics repository under permissive licensing.

1. **NASA Blue Marble 2K Equirectangular Surface Texture (earth_surface_2k.jpg):** True-color global mosaic composited from observations by the MODIS instrument on NASA's Terra and Aqua satellites. Resolution: 2048 × 1024 pixels (512,606 bytes). Applied directly as meshStandardMaterial.map on Earth's 3D sphere in both the Title Screen 3D Hero Visual (HeroOrbitalVisual.tsx:137) and the Mission Control Orbit Map (PlanetaryScene3D.tsx:137).
2. **NASA Blue Marble Ocean Specular / Reflectance Mask (earth_specular_2k.jpg):** Grayscale land/water mask derived from MODIS terrestrial classification (223,421 bytes). Oceans are white (high specularity) and continental landmasses are black (matte). Mapped to roughnessMap (PlanetaryScene3D.tsx:140, HeroOrbitalVisual.tsx:135) so that directional sunlight creates authentic glint off oceans while land remains diffuse.
3. **NASA SRTM Topographic Surface Normal Map (earth_normal_2k.jpg):** Tangent-space surface normal vector map derived from the Shuttle Radar Topography Mission (SRTM30) digital elevation model (336,774 bytes). Applied as normalMap with normalScale = Vector2(0.6, 0.6) (PlanetaryScene3D.tsx:138-139), producing authentic physical bump lighting across the Himalayas, Andes, and oceanic trenches.
4. **NASA/NOAA Meteorological Satellite Cloud Layer Alpha Composite (earth_clouds_4k.png):** Authentic 4096 × 2048 equirectangular cloud coverage map compiled from GOES and polar weather satellites (5,033,486 bytes). Rendered as a separate concentric 3D sphere at radius = 1.012 × Earth Radius with opacity = 0.28 (PlanetaryScene3D.tsx:147-150). The cloud layer rotates at 0.041 rad/s while the surface rotates at 0.032 rad/s, modeling real atmospheric differential super-rotation.

B. NASA Planetary Fact Sheet Physical Constants (NSSDC / GSFC)

Located in src/data/nasa/planets/planetaryData.ts:3-85 and served via src/app/api/nasa/planetary/route.ts. Compiled from Dr. David R. Williams' canonical Planetary Fact Sheet repository at NASA Goddard Space Flight Center:

- **The Moon:** Mass 7.342 × 10^{22} kg (0.0123 Earths); Mean radius 1,737.4 km; Surface gravity 1.62 m/s² (0.166 g); Escape velocity 2.38 km/s; Orbit semi-major axis 384,400 km; Orbital period 27.3217 days; Surface-boundary exosphere mixing ratios: Helium-4 (40,000 atoms/cm³), Neon (40,000), Hydrogen (35,000), Argon-40 (30,000); Diurnal thermal range -130°C to +120°C (Permanently Shadowed Regions dip to -246°C / 27 K); Radiation dose ~60 µSv/hr (LRO CRaTER & Apollo EASEP ground truth).
- **Mars:** Mass 6.417 × 10^{23} kg (0.107 Earths); Mean radius 3,389.5 km; Surface gravity 3.72 m/s² (0.38 g); Escape velocity 5.03 km/s; Average solar distance 225,000,000 km; Orbital period 686.98 days; Atmospheric composition: 95.32% CO■, 2.6% N■, 1.9% Ar, 0.16% O■, 0.08% CO; Thermal extremes -125°C to +20°C (mean -63°C); Surface radiation dose ~250 mSv/year (MSL Curiosity RAD verified).
- **Earth:** Mass 5.972 × 10^{2}■ kg; Mean radius 6,371.0 km; Surface gravity 9.807 m/s²; Escape velocity 11.186 km/s; Average distance 149,597,870 km; Orbital period 365.256 days; Atmosphere: 78.08% N■, 20.95% O■, 0.93% Ar, 0.04% CO■; Thermal extremes -89.2°C to +56.7°C; Van Allen radiation belt proton/electron shielding.

C. Authentic NASA LRO Scientific Instrument Payload Suite (NASA PDS)

Located in src/data/nasa/instruments/lroInstruments.ts:3-190 and integrated into spacecraft construction in src/data/components.ts:110-212. Each instrument incorporates verified engineering specs, principal investigators, and official NASA Planetary Data System (PDS) archive identifiers:

1. **LOLA (Lunar Orbiter Laser Altimeter):** PDS ID LRO-L-LOLA-3-RDR-V1.0. PI: Dr. David E. Smith (NASA GSFC). Real Mass: 11.2 kg; Real Power: 30.8 W. Physics: 5-beam Nd:YAG 1064 nm pulsed laser rangefinder at 28 pulses/sec. Measures time-of-flight to generate millimetric 3D digital elevation models.
2. **LROC (Lunar Reconnaissance Orbiter Camera):** PDS ID LRO-L-LROC-2-EDR-V1.0. PI: Dr. Mark Robinson (Arizona State University). Real Mass: 19.2 kg; Real Power: 34.0 W. Physics: Dual Narrow Angle Cameras (NAC, 0.5 m/pixel panchromatic) and one Wide Angle Camera (WAC, 100 m/pixel 7-band multispectral).

3. **Diviner (Diviner Lunar Radiometer Experiment - DLRE):** PDS ID LRO-L-DLRE-4-RDR-V1.0. PI: Dr. David A. Paige (UCLA). Real Mass: 13.0 kg; Real Power: 24.7 W. Physics: 9-channel infrared filter radiometer (0.3 to >200 µm). Discovered cryogenic polar cold traps below 30 K (-243°C).
4. **Mini-RF (Miniature Radio Frequency Synthetic Aperture Radar):** PDS ID LRO-L-MRFLRO-4-CDR-V1.0. PI: Dr. Ben Bussey (JHU/APL). Real Mass: 15.6 kg; Real Power: 90.0 W. Physics: Dual-frequency S-band (2.38 GHz) and X-band (7.14 GHz) radar measuring Circular Polarization Ratio (CPR) for buried water ice.
5. **LAMP (Lyman-Alpha Mapping Project):** PDS ID LRO-L-LAMP-2-EDR-V1.0. PI: Dr. Kurt Retherford (SwRI). Real Mass: 6.1 kg; Real Power: 4.5 W. Physics: Far-UV imaging spectrograph (57–196 nm) utilizing interplanetary hydrogen Lyman-alpha starlight glow to map frost.
6. **CRaTER (Cosmic Ray Telescope for the Effects of Radiation):** PDS ID LRO-L-CRAT-2-EDR-V1.0. PI: Dr. Nathan Schwadron (UNH). Real Mass: 5.4 kg; Real Power: 7.3 W. Physics: Silicon solid-state detector stack with tissue-equivalent plastic (TEP) measuring linear energy transfer (LET) cosmic ray spectra.

D. NASA Historical Robotic Spacecraft & Mission Science Profiles (NSSDCA)

Located in src/data/nasa/missions/missionProfiles.ts:3-124 and served via src/app/api/nasa/missions/route.ts. Maintains canonical profiles from the NASA Space Science Data Coordinated Archive:

- **LRO:** Launch June 18, 2009 (SLC-41, Atlas V 401); Destination: Lunar Polar Orbit (50 km altitude); Status: Operational; Data Return: 1,450.0 Terabytes downlinked to PDS; Primary Goals: Geodetic grid, radiation mapping, polar water ice.
- **Apollo 11:** Launch July 16, 1969 (LC-39A, Saturn V SA-506); Landing: Mare Tranquillitatis (0.674° N, 23.473° E); Data Return: 0.08 TB digitized telemetry; Pristine sample return: 21.55 kg lunar rocks; Experiments: EASEP (PSEP, LRRR, SWC).
- **Mars 2020 (Perseverance & Ingenuity):** Launch July 30, 2020 (SLC-41, Atlas V 541); Landing: Jezero Crater (18.38° N, 77.58° E); Data Return: 110.0 TB; Primary Goals: Biosignatures, core caching for Mars Sample Return, in-situ MOXIE oxygen generation.
- **OSIRIS-REx:** Launch Sept 8, 2016 (SLC-41, Atlas V 411); Destination: Asteroid 101955 Bennu; Sample Return: 121.6 grams delivered to Earth Sept 24, 2023; Data Return: 45.0 TB; Extended Mission: OSIRIS-APEX to Apophis.

E. Satellite Ephemeris, DSN Telemetry Protocols & Heliophysics Models

1. **Orbital Ephemeris State Vectors (CelesTrak / NASA Tracking Network):** Defined in src/data/satellites/satelliteNetwork.ts:30-166. Contains authentic orbital inclinations, altitudes, eccentricities, periods, RAAN, and velocities for **ISS** (418 km, 51.64°, 7.66 km/s, NORAD 25544), **Hubble Space Telescope** (535 km, 28.47°, 7.59 km/s, NORAD 20580), **Landsat 9** (705 km Sun-Synchronous, 98.2°, 7.50 km/s, NORAD 49260), and **NOAA-20 / JPSS-1** (824 km Polar LEO, 98.7°, 7.44 km/s, NORAD 43013). Orbital points are computed via exact Keplerian-to-Cartesian transformation (calculateOrbitPoint) and rendered as real-time 3D orbits with interactive telemetry modals.
2. **NASA Deep Space Network (DSN) Telemetry Protocols (JPL D-16838):** Informs 8.4 GHz X-band carrier lock telemetry in the Orbit Map (OrbitMapCanvas.tsx:163-174) and grounds in-flight crisis Event evt-comm-drop (events.ts:47-87) simulating signal carrier loss, antenna deflection, and helical recovery scanning across Goldstone, Madrid, and Canberra.
3. **NASA DONKI / NOAA SWPC Solar Flare & Proton Flux Models:** Informs in-flight crisis Event evt-solar-storm (events.ts:4-45), modeling relativistic proton bombardment from Class-X solar flares, single-event upsets (SEUs) in avionics, and authentic spacecraft Safe Mode procedures.

4. Feature Mapping Matrix

This matrix details exactly which user-facing screen, interactive tool, or simulation canvas consumes and displays each confirmed NASA dataset.

NASA Dataset	Project Component / Screen	Exact UI / Simulation Function
Blue Marble Surface, Specular & Normal (MODIS/SRTM)	PlanetaryScene3D.tsx HeroOrbitalVisual.tsx	3D Earth sphere rendering with true-color surface, ocean glint specular reflections, and normal elevation bump lighting under directional sunlight.
Real Atmospheric Cloud Layer (GOES/NOAA)	PlanetaryScene3D.tsx HeroOrbitalVisual.tsx	Concentric 3D cloud sphere rotating at differential speed (+28% faster than surface) to simulate real atmospheric super-rotation.
Planetary Physical Constants (Moon, Mars, Earth)	DataExplorerScreen.tsx MissionBriefingScreen.tsx	Interactive Planetary Properties explorer tab displaying authentic mass, radius, surface gravity, escape velocity, and atmospheric profiles.
LRO LOLA Laser Altimeter GDR (PDS)	SpacecraftBuilderScreen.tsx SpacecraftBlueprintCanvas.tsx	Selectable scientific instrument component (nasa-inst-lola), animated CAD laser rangefinder aperture, and 3D Topographic synergy bonuses.
LRO LROC Multispectral Camera Archive (PDS)	SpacecraftBuilderScreen.tsx SpacecraftBlueprintCanvas.tsx	Selectable instrument (nasa-inst-lroc) with authentic 19.2 kg mass and 34 W draw; renders animated dual telephoto lens in CAD schematic.
LRO Diviner Thermal Data (PDS)	SpacecraftBuilderScreen.tsx EventModal.tsx DebriefScreen.tsx	Component nasa-inst-diviner; drives polar volatile anomaly event and post-mission debrief analysis on 25 K polar cold traps.
LRO Mini-RF SAR Polar Ice Data (PDS)	SpacecraftBuilderScreen.tsx SpacecraftBlueprintCanvas.tsx	Component nasa-inst-minirf; renders animated microwave radar pulse waves and unlocks cryogenic volatile ice detection synergies.
LRO LAMP Far-UV Spectrograph (PDS)	SpacecraftBuilderScreen.tsx DataExplorerScreen.tsx	Component nasa-inst-lamp; models Lyman-alpha starlight reflection and water frost diagnostic absorption edges in permanently shadowed craters.
LRO CRaTER Radiation Archive (PDS)	SpacecraftBuilderScreen.tsx EventModal.tsx	Component nasa-inst-crater; grants -8% mission flight risk reduction and provides historical context for coronal mass ejection crisis events.
Authentic NASA LRO Preset Build	gameStore.ts:391-408 SpacecraftBuilderScreen.tsx	Dedicated 'REAL LRO PRESET (NASA PDS)' button equipping all 4 primary LRO payloads to trigger the authentic +40 Science suite bonus.
NASA Historical Robotic Missions Catalog	DataExplorerScreen.tsx api/nasa/missions/route.ts	Dedicated 'Robotic Missions' browser displaying launch vehicles, Cape Canaveral pad locations, PDS science volumes (TB), and official mission goals.
Satellite Ephemeris (ISS, HST, Landsat 9, NOAA-20)	HeroOrbitalVisual.tsx satelliteNetwork.ts	Renders 3D orbital trajectory loops and custom 3D CAD silhouettes; clicking any spacecraft opens live orbital telemetry inspector modal.
Deep Space Network Carrier Standards	OrbitMapCanvas.tsx EventModal.tsx	Dynamic radio beam connecting Earth ground stations to probe with 'DSN CARRIER: 8.4 GHz LOCK'; triggers DSN carrier drop emergency choices.
NASA Provenance Registry & Verification	DataSourcesScreen.tsx DataProvenanceModal.tsx	Searchable transparency table with official NASA source links, DOIs, retrieved dates, and strict separation between real data and game simulation.

5. Source-Code Evidence and Citations

Every confirmed NASA data product was verified directly in source code. The following citations document exact file locations and code line numbers:

Target Data Product	Source File Path	Lines	Code Symbol / Variable / Implementation
Blue Marble Surface Texture	src/lib/planetTextures.ts	18, 70	const EARTH_SURFACE = '/textures/earth_surface_2k.jpg'; loadEarthTexture(..., true)
Blue Marble Specular Mask	src/lib/planetTextures.ts	19, 71	const EARTH_SPECULAR = '/textures/earth_specular_2k.jpg'; roughnessMap
SRTM Elevation Normal Map	src/lib/planetTextures.ts	20, 72	const EARTH_NORMAL = '/textures/earth_normal_2k.jpg'; normalMap
Real Atmospheric Cloud Map	src/lib/planetTextures.ts	21, 73	const EARTH_CLOUDS = '/textures/earth_clouds_4k.png'; alphaMap
Earth Mesh Texture Binding	src/components/canvas/PlanetaryScene3D.tsx	134-150	meshStandardMaterial: map, normalMap, roughnessMap, alphaMap (clouds)
Central Earth Hero Rendering	src/components/canvas/HeroOrbitalVisual.tsx	128-152	CentralEarth3D(): sphereGeometry(0.82), map, cloud sphere(0.82*1.012)
NASA Moon Physical Constants	src/data/nasa/planets/planetaryData.ts	4-30	REAL_PLANETARY_DATA.moon: massKg='7.342e22', gravity=1.62, escapeVel=2.38
NASA Mars Physical Constants	src/data/nasa/planets/planetaryData.ts	31-57	REAL_PLANETARY_DATA.mars: massKg='6.417e23', gravity=3.72, escapeVel=5.03
NASA Earth Physical Constants	src/data/nasa/planets/planetaryData.ts	58-84	REAL_PLANETARY_DATA.earth: massKg='5.972e24', gravity=9.807, escapeVel=11.186
Planetary API Route	src/app/api/nasa/planetary/route.ts	1-19	GET(req): returns REAL_PLANETARY_DATA[target] with X-NASA-Data-Source
LRO LROC Instrument Data	src/data/nasa/instruments/lroInstruments.ts	3-34	REAL_LRO_INSTRUMENTS[0]: 'nasa-inst-lroc', massKg=19.2, powerW=34.0, PDS EDR
LRO LOLA Instrument Data	src/data/nasa/instruments/lroInstruments.ts	36-65	REAL_LRO_INSTRUMENTS[1]: 'nasa-inst-lola', massKg=11.2, powerW=30.8, PDS RDR
LRO Diviner Instrument Data	src/data/nasa/instruments/lroInstruments.ts	67-96	REAL_LRO_INSTRUMENTS[2]: 'nasa-inst-diviner', massKg=13.0, powerW=24.7, 30K PSR
LRO Mini-RF SAR Instrument Data	src/data/nasa/instruments/lroInstruments.ts	98-127	REAL_LRO_INSTRUMENTS[3]: 'nasa-inst-minirf', massKg=15.6, powerW=90.0, CPR
LRO LAMP Spectrograph Data	src/data/nasa/instruments/lroInstruments.ts	129-158	REAL_LRO_INSTRUMENTS[4]: 'nasa-inst-lamp', massKg=6.1, powerW=4.5, Lyman-alpha
LRO CRaTER Radiation Data	src/data/nasa/instruments/lroInstruments.ts	160-189	REAL_LRO_INSTRUMENTS[5]: 'nasa-inst-crater', massKg=5.4, powerW=7.3, LET Spectra
LRO Instruments Spacecraft Adapter	src/data/components.ts	111-212	SPACECRAFT_COMPONENTS: realMassKg, realPowerW, isRealNasaInstrument=true
LRO Science Preset Generator	src/store/gameStore.ts	391-408	loadLroBuild(): equips lroc, lola, diviner, minirf (+40 science synergy)
Robotic Missions Master Catalog	src/data/nasa/missions/missionProfiles.ts	3-123	REAL_NASA_MISSIONS: LRO (1450 TB), Apollo 11 (0.08 TB), M2020 (110 TB), OREx (45 TB)
Satellite Network Orbital TLEs	src/data/satellites/satelliteNetwork.ts	30-166	REAL_SATELLITE_NETWORK: ISS (25544), HST (20580), L9 (49260), N20 (43013)
Keplerian Orbit Calculation	src/data/satellites/satelliteNetwork.ts	172-199	calculateOrbitPoint(sat, theta, scale): orbital plane to ECI transformation
Coronal Mass Ejection Crisis Event	src/data/events.ts	5-44	evt-solar-storm: cites NASA Heliophysics DONKI & LRO CRaTER PDS archive

DSN Carrier Loss Crisis Event	src/data/events.ts	47-86	evt-comm-drop: cites NASA Deep Space Network Telemetry Standard (JPL D-16838)
Polar Volatiles Discovery Event	src/data/events.ts	131-169	evt-anomaly-discovery: cites LRO Diviner & Mini-RF PSR Volatiles Report
NASA Provenance Registry	src/data/nasa/metadata/provenanceCatalog.ts	3-164	NASA_PROVENANCE_CATALOG: 14 catalog records with source URLs, DOIs, and PDS nodes

6. Official NASA Documentation & Archive Links

Every identified dataset is traceable to an official public NASA data repository, PDS node, or mission center. The following links provide official documentation:

Target Scientific Domain	Official NASA / Agency Repository URL	Archive Node / Division
NASA Blue Marble Global Imagery	`https://visibleearth.nasa.gov/collection/1484/blue-marble`	NASA Earth Observatory / GSFC
NASA SRTM Radar Topography	`https://www2.jpl.nasa.gov/srtm/`	NASA Jet Propulsion Laboratory (JPL)
NASA Earthdata Environmental Portal	`https://www.earthdata.nasa.gov/`	NASA ESDIS Project
NASA Planetary Fact Sheet - Moon	`https://nssdc.gsfc.nasa.gov/planetary/factsheet/moonfact.html`	NASA Goddard Space Flight Center (NSSDC)
NASA Planetary Fact Sheet - Mars	`https://nssdc.gsfc.nasa.gov/planetary/factsheet/marsfact.html`	NASA Goddard Space Flight Center (NSSDC)
NASA Planetary Fact Sheet - Earth	`https://nssdc.gsfc.nasa.gov/planetary/factsheet/earthfact.html`	NASA Goddard Space Flight Center (NSSDC)
LRO LOLA Laser Altimetry Archive	`https://pds-geosciences.wustl.edu/missions/lro/lola.htm`	NASA PDS Geosciences Node (WUSTL)
LRO LROC Camera Public Node	`https://lroc.sese.asu.edu/`	NASA PDS Imaging Node / Arizona State Univ
LRO Diviner Thermal Radiometer	`https://www.diviner.ucla.edu/`	NASA PDS Geosciences Node / UCLA
LRO Mini-RF Radar Polar Science	`https://pds-geosciences.wustl.edu/missions/lro/minirf.htm`	NASA PDS Geosciences Node / JHU-APL
LRO LAMP UV Spectrograph Archive	`https://pds-atmospheres.nmsu.edu/data_and_services/atmospheres_data/LRO/lamp.html`	NASA PDS Atmospheres Node (NMSU)
LRO CRaTER Cosmic Ray Dosimetry	`https://pds-ppi.igpp.ucla.edu/mission/LRO/CRAT/`	NASA PDS PPI Node / UCLA / UNH
NASA NSSDCA Master Missions Catalog	`https://nssdc.gsfc.nasa.gov/`	NASA Goddard Space Flight Center
NASA Deep Space Network (DSN) Ops	`https://eyes.nasa.gov/dsn/dsn.html`	NASA Jet Propulsion Laboratory (JPL)
NASA Space Weather DONKI Portal	`https://kauai.ccmc.gsfc.nasa.gov/DONKI/`	NASA Community Coordinated Modeling Center
USGS / NASA Landsat-9 Mission	`https://landsat.gsfc.nasa.gov/satellites/landsat-9/`	NASA GSFC / USGS EROS Center
NOAA-20 / NASA JPSS Mission	`https://www.nesdis.noaa.gov/our-satellites/currently-flying/joint-polar-satellite-system`	NOAA NESDIS / NASA Joint Program
International Space Station (ISS)	`https://www.nasa.gov/international-space-station/`	NASA Space Operations Directorate
Hubble Space Telescope Science Node	`https://hubblesite.org/`	NASA GSFC / Space Telescope Science Inst

7. Suspected, Simulated, or Unverified Data (Clearly Marked as Unconfirmed)

To maintain rigorous scientific audit standards, the following items were identified in the codebase and investigated, but are **explicitly designated as unconfirmed, simulated, procedural, or unused**:

Unverified / Procedural Item	Source Location	Audit Finding & Technical Explanation
Unused Asset: earth_topology.png	`public/textures/earth_topology.png`	**UNCONFIRMED / UNUSED ASSET:** A 378,243-byte image exists in the static asset directory. Grep analysis across all TypeScript, CSS, and JSX source files reveals **zero occurrences**. This file is completely unreferenced by the runtime rendering pipeline.

Procedural Lunar Surface Textures	`src/lib/planetTextures.ts:83-250`	**PROCEDURAL ARTWORK (NOT RAW NASA RASTER):** While provenanceCatalog.ts:105-113 cites the LRO WAC Global Mosaic, the runtime texture is procedurally painted via HTML5 2D Canvas routines (drawMare, drawCraterWithRays). No real NASA photographic raster is bundled for the Moon.
Procedural Mars Surface Textures	`src/lib/planetTextures.ts:252-335`	**PROCEDURAL ARTWORK (NOT RAW NASA RASTER):** While provenanceCatalog.ts:115-123 cites MGS MOLA and MRO MARCI, the Mars surface texture is procedurally generated with canvas gradient circles representing Olympus Mons and Valles Marineris. It is an artistic approximation, not downloaded raw MGS MOLA MEGDR data.
Fictional Spacecraft & Launcher Specs	`src/data/launchers.ts:1-38` `src/data/components.ts:1-108`	**SIMULATED GAME BALANCING:** Generic components (Carbon-Composite Bus, Titanium Bus, Battery Banks, Thrusters) and launch vehicles (AeroLift-1, Atlas-Titan V, Nova Heavy) use fictional cost, mass, and thrust numbers engineered for gameplay progression, not genuine NASA engineering telemetry.
Procedural Web Audio Engine	`src/lib/sound.ts:1-871`	**PROCEDURAL SYNTHESIS (NO NASA AUDIO RECORDINGS):** All launch acoustics (rumble, shear, turbopump hum, countdown beeps, flight director chords) are synthesized procedurally via Web Audio API oscillators and noise buffers. No authentic NASA air-to-ground mission audio recordings are used.

8. Final Audit Count of Data Products & Datasets

To avoid double-counting multi-format representations or related instruments, the final audit count classifies items into unambiguous scientific categories:

Classification Category	Distinct Count	Reconciliation & Non-Duplication Rule
Confirmed NASA Cartographic Image Products	**4**	1. Blue Marble Surface (2K RGB); 2. Blue Marble Specular Mask; 3. SRTM Normal Map; 4. GOES Cloud Map. (Each is an independent physical file in /public/textures/).
Confirmed NASA Planetary Science Datasets	**1**	NASA GSFC/NSSDC Planetary Constants (covers Moon, Mars, Earth physical parameters counted as a single unified astronomical catalog).
Confirmed NASA LRO Scientific Instrument Datasets	**6**	LOLA, LROC, Diviner, Mini-RF, LAMP, and CRaTER. (Each represents an independent instrument dataset downlinked to distinct NASA PDS nodes).
Confirmed NASA Historical Missions Catalog	**1**	NASA Space Science Data Coordinated Archive Master Catalog (encompasses LRO, Apollo 11, Mars 2020, OSIRIS-REx profiles).
Confirmed Space Surveillance Ephemeris & DSN Feeds	**3**	1. CelesTrak/NASA Tracking Network Satellite Ephemeris (ISS, HST, Landsat 9, NOAA-20); 2. DSN Telemetry Standards; 3. DONKI/SWPC Space Weather Flare Models.
TOTAL CONFIRMED SPACE SCIENCE DATA SOURCES	**15**	**12 Pure NASA Data Products + 3 Allied Space Science / Orbit Surveillance Datasets**
Suspected, Unused, or Procedural Sources	**5**	1. earth_topology.png (unused asset); 2. Procedural Moon Canvas; 3. Procedural Mars Canvas; 4. Fictional launcher/bus specs; 5. Procedural Web Audio engine.

Audit Conclusion: The *Beyond-Earth* software repository demonstrates a genuine, verified integration of NASA data products. It rigorously implements 4 real cartographic satellite rasters, 1 planetary science physical constant suite, 6 authentic Lunar Reconnaissance Orbiter PDS instrument datasets, 1 historical robotic missions catalog, and 3 allied space ephemeris and telemetry protocols. All simulated mechanics and procedural graphics are accurately differentiated from genuine scientific data. No spurious or fabricated NASA datasets were detected in the codebase.